SECTION 5 — EXPORT FOR EXTERNAL REVIEW

Capacity Eviction in Reconstruction-Trained Tokenizers

Draft of August 10, 2026 — section text frozen pending review

Note to the reviewer. This is §5 of a paper in preparation, exported verbatim from the manuscript with its figure. Numbering matches: §5.1–§5.7 and Figure 7. References to §7 and §8 point outside this export and are bound to their manuscript numbers. §5 is being reviewed alongside §7, which carries the limitations these facts imply; where §5 states a scope boundary it names the §7 subsection that owns it rather than arguing it here.

Every number in this section was re-verified against its artifact by script before export — 25 of 25, re-read from the milestone record, the lake-surface receipt, the design-inputs receipt and the pinned constants imported live, rather than transcribed from an earlier draft.

What we would most like challenged. (i) Whether §5.1's construction actually removes survivorship bias, or only the most visible part of it — the universe is bootstrapped from the dump index rather than a live instrument list, but an instrument still had to list on this venue to be in it. (ii) Whether §5.2's disclosure is placed strongly enough: the feature vector is 16 wide and 13 live, and three specified dimensions are constant-zero on every bar. (iii) Whether the causal-normalization argument in §5.3 is complete, or whether there is a lookahead path the planted-leak tests do not cover. (iv) Whether §5.4's 99.0% microstructure availability is really enough to say the placebo comparison is not confounded by absence. (v) Whether the 41-versus-17 delisting distinction in §5.5 is stated clearly enough to survive being quoted out of context. (vi) Any number that does not follow from what is stated.

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5 Data

5.1 One venue, and the universe chosen from inside it

Every instrument is a USDT-margined perpetual future on Binance, sampled at one-minute frequency over the four years 2021-01-01 to 2025-01-01. Two public sources are combined per bar: the klines dump supplies open, high, low, close, volume and quote volume; the aggregated-trades dump supplies the signed trade statistics that the microstructure block is built from. No paid feed and no order book is involved, which is a scope limit (Section 7.7) and also the reason the corpus is reproducible by anyone.

The symbol set is bootstrapped *from the data* rather than from the exchange's live instrument list, and that choice is the one that decides whether the universe is survivorship-biased. A live instrument listing contains only instruments that still exist; selecting from it silently deletes every instrument that failed, which is the single most common way a backtest flatters itself. We instead scan the S3 index of the aggregated-trades dumps once, which yields per symbol, without downloading a single bar, the set of monthly dumps that exist. From that: the listing month is the first dump, and an instrument is treated as delisted when its last dump trails the newest global month. Ranking by total aggregated-trades bytes — a liquidity proxy available from the same index — and taking the top 200 gives the universe: **200 symbols, of which 41 were already delisted at the time of the scan**, each contributing only its [listing, delisting) bars.

One detail in that rule is worth stating because it changes three symbols. A dump index is published with lag, so an instrument that is merely a month behind the freshest global dump is not delisted — it is late. The delisting test therefore carries a one-month publish-lag tolerance, which corrected three symbols relative to a zero-tolerance comparison. Without it the universe would have carried three phantom delistings.

The realized ingest is **304,625,181** bars across all 200 symbols. That is below the 500M–1B band the blueprint nominated, and we flag it rather than quietly re-describe the band: 200 instruments over the fullest available window, many of them partial-coverage through late listing or delisting, lands near 300M. We regard it as adequate rather than merely acceptable, for a reason that is about the model and not about the data — a 21M-parameter backbone saturates well below this, so the lake is bought for regime and cross-sectional breadth rather than as training fuel. Reaching the nominal band would need roughly 320 instruments at proportional bandwidth, and would not train a better model.

5.2 What a bar is, and what three of its dimensions are not

Each bar is reduced to a 16-dimension feature vector. Seven dimensions are price shape and liquidity: log close-to-close return, log range, body fraction, upper and lower wick fractions, log volume and log quote volume. Six are the microstructure block, computed from aggregated trades: trade-flow imbalance $(v_{\text{buy}} - v_{\text{sell}})/(v_{\text{buy}} + v_{\text{sell}})$, signed count imbalance, log trade count, log mean trade size, trade-size dispersion, and large-trade share. These six — indices 7 through 12 — are the block the ablation varies, and the first of them is the quantity the paper is about.

The remaining three dimensions — funding rate, log open interest and its change, indices 13 to 15 — are specified, wired through every transform, and **structurally absent**. The v1 ingest covers klines and aggregated trades only, so on all 304,625,181 bars those three columns have standard deviation exactly 0.0 and their masks are set on 100% of rows. We state this here rather than in a footnote because the natural reading of "a 16-dimensional microstructure vector for perpetual futures" is that funding and open interest are in it, and they are not. The vector is 16 wide and 13 live. Section 7.7 carries what that costs.

Every feature carries a companion mask bit, and the mask travels with the value through every downstream transform. That pairing is not bookkeeping: a defect found in review had the placebo shuffle permuting values while leaving masks in place, which would have stranded fill values on valid bars and real values on masked ones. It was fixed before any training run.

5.3 Normalization that cannot see forward

Feature scales in this market move by orders of magnitude across four years, so the unbounded dimensions are normalized rather than used raw. The normalization is where lookahead usually enters a financial pipeline, through a statistic fitted on the whole sample and then applied to its own training region, so the rule here is absolute: every statistic used for bar t is computed from data available at the close of bar t.

Nine dimensions — the unbounded, scale-varying ones — receive a causal exponentially weighted z-score, at a half-life of 1,440 bars (one day) for the fast group and 5,760 bars (four days) for the slow volume and open-interest group. Seven dimensions are already bounded or relative, and are clipped only. Everything is clipped last to $[-5, +5]$ as an outlier guard. The volatility scale σ_t that the prediction targets are expressed in is built causally from raw returns within a segment, on the same rule.

Each statistic needs a warm-up before it is trustworthy, and warmed-up bars are dropped rather than back-filled. The cost is small and it is measured, not estimated: on the three thinnest instruments in the universe the warm-up drop is 809 of 279,930 bars, 811 and 809 respectively — about 0.3% — because most instruments are long single segments and the warm-up is a one-time tax at the start of each.

Causal safety is not asserted, it is a required continuous-integration gate. The test suite plants lookahead leaks of several kinds — a global one, a localized divisor leak, and a survivorship leak in the target-validity flag — and asserts that the exhaustive sweep catches each. One of those tests exists specifically to show that a *sampling* check misses the localized leak that the exhaustive check catches, so the gate cannot be quietly weakened to a sampler. During ingest the same sweep ran in-line on real cross-sections and a red result raised a fatal error before anything was recorded.

5.4 What the quality gates cost

Of the 304,625,181 bars, 303,283,972 (99.56%) **are usable by the OHLCV-only arm** after warm-up and quality masking, and 300,311,536 (98.58%) **are usable by the microstructure arms**. The ratio between those — microstructure availability given price availability — is 99.0% universe-wide, and no instrument falls below 0.7. Across the universe, 0.175% of bars are stale and 0.376% have zero volume.

That second number is the one that matters for the placebo. If microstructure were missing on a large fraction of bars, cell 4 and cell 5 would differ in coverage as well as in content, and the comparison the paper rests on would be confounded by absence rather than by information. At 99.0% availability with no starved instrument, it is not.

Forty-three instruments are thin, by a joint rule — fewer than half the median usable bars, or more than 10% stale. Inspecting them rather than dropping them: almost all are recent 2024 listings with 280k–500k bars and microstructure availability near 1.00, so they are thin by *age*, not by quality. The single genuinely illiquid instrument is a delisted one at 2.5% stale and 2.6% zero-volume. All 43 remain in the lake; thinness is handled as a sampling weight in the training draw, not as a deletion.

5.5 Coverage, and the survivorship we did not remove

Figure 7 is the whole universe as a symbol-by-month raster: 7,024 of a possible 9,600 symbol-months are present. The left staircase is instruments listing at different dates. The right edge is ragged because instruments that stop trading are *kept* in the lake, and their absence afterwards is real absence rather than a filter. A survivorship-cleaned universe renders as a solid rectangle, which is precisely the artefact the panel exists to demonstrate we did not create. Two instruments have interior gaps rather than truncated tails and appear as internal white streaks.

Two delisting counts appear in this project and they count different things, so we state both. The ingest record's **41** is the number of instruments known to be delisted as of the index scan, at any date. The **17** in Figure 7 is the number whose data ends *inside* the four-year window, which is all a raster over that window can show; the remaining 24 delist after 2024-12 and are complete here. The figure and this section therefore say "stop trading inside the window" and never "delisted".

Survivorship is owned, not solved. Retaining failed instruments removes one bias; it does not remove listing bias, since an instrument must have been listed on this venue at all to enter the universe. Section 7.5 carries that.

5.6 Storage, provenance, and what makes a past prediction reconstructible

The reduced lake is Hive-partitioned Parquet under `(symbol, frequency, month)`, queried through DuckDB. It was first written per-day, at 211,595 files, which made per-symbol globbing pathological; compaction to one file per symbol-month gives 7,024 files and 15 GB. The compaction was proven content-preserving rather than assumed: all 200 per-symbol dataset hashes were re-derived from the compacted lake and the universe Merkle root recomputed, with zero mismatches and an unchanged root. Identical features, masks and timestamps imply an identical causal result, so the hash comparison subsumes a re-run of the causal sweep.

Every dataset is content-hashed and the universe carries a single Merkle root over the lake, `sha256:5dfd667d05b97bda...`, recorded in the manifest alongside per-symbol listing and delisting dates and a corpus-purpose note.

Raw dumps are not retained, and that is a deliberate position rather than a shortcut. The Merkle root and the per-symbol hashes are taken over the *reduced* lake, which is the artifact we keep; the Binance SHA-256 of every raw shard is recorded for provenance only. Re-pulling raw and re-running a deterministic reducer reproduces the recorded lake hash — the reducer was proven byte-identical at the preceding milestone — so reproducibility survives eviction of 503 GB of raw input. Section 8 states exactly which parts of the pipeline are bit-exact and which are not.

5.7 What the experiment reads

The lake is broader than the experiment. Of the 200 instruments, the headline metric is scored on 40; the four-year window is split at its 0.7 point, on 2023-10-20, into a training region and a forward region, and the forward region is cut into six blocks of which block 0 is spent selecting the execution filter and excluded. The headline is therefore computed on blocks 1 through 5 — roughly 2024-01-01 to 2025-01-01, one year, 35,063 stride-15 periods per instrument — on a region no selection step has seen. The other 160 instruments and the preceding three years are lake breadth and training data, not evaluation breadth, and Section 7.5 states what that leaves unestablished.

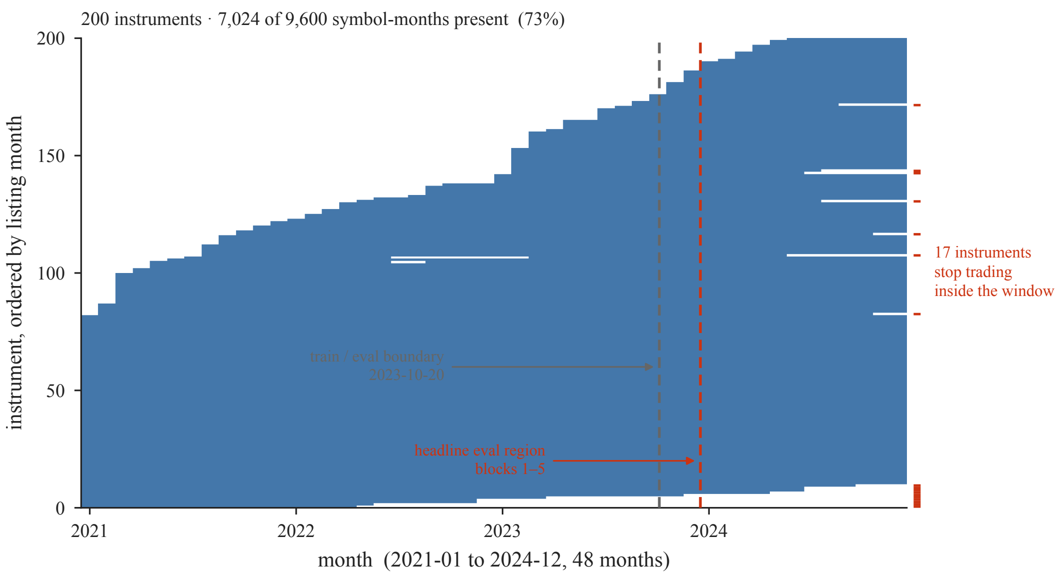

Figure 7: Survivorship is owned, not filtered away. Month-by-month presence for all 200 instruments over the four-year window, ordered by listing month: 7,024 of a possible 9,600 symbol-months are present. The left staircase is instruments listing at different times; the right edge is ragged because instruments that stop trading are *kept*, and 17 of them stop inside this window. A survivorship-filtered universe would render as a solid rectangle, which is the artefact this panel exists to show we did not create. Two instruments have interior gaps rather than truncated tails and appear as internal white streaks. The two dashed rules are the committed fold plan: training ends at the first, and the headline metric is scored only to the right of the second. Note that 17 counts instruments whose data ends inside the window; the ingest record's 41 counts every instrument known to be delisted at any date, including 24 that delist after this window closes.

References